Data centers and the cost of the AI build out

What the growth of AI infrastructure could mean for your electricity bill and your community

Titans Forge LLC | Updated September 26, 2026

A new data center can bring investment and useful computing capacity. It can also require new power lines, substations and generating capacity. For the people living nearby, the practical question is who pays for that work, especially if the project grows more slowly than promised.

What our report finds

Demand for AI infrastructure can be real while individual projects become too expensive, arrive late or disappoint investors. We examine the connections between computing demand, equipment supply, financing and local electricity costs. A useful technology does not guarantee a sound project.

What the evidence adds

The IEA reports 17% growth in global data-center electricity consumption in 2025 and projects 950 TWh in 2030. That is a global projection, not a forecast for your town. Local supply and the way costs are recovered determine the effect on bills. [1]

Virginia illustrates the importance of those rules. Its SCC describes special rates and contract protections for qualifying Dominion large-load customers, including a 14-year obligation for new customers contracting from January 2027. The aim is to reduce costs being shifted to other users. [2]

Three things worth separating

- Announced capacity and operating capacity. A request for electricity is not proof that a facility will use it.
- A useful industry and a worthwhile investment. Demand growth can coexist with weak project economics.
- A larger electricity system and a larger household bill. Contracts, regulation and spare capacity affect who bears the cost.

Questions to ask about a local project

What will be built, when will it connect, and how much demand is backed by a contract? Ask which upgrades serve the project directly and which benefit the wider grid. Then ask what happens to the costs if the customer delays, scales back or leaves.

Look for the approved tariff and the actual obligations. Minimum payments, collateral and exit provisions matter more than a general promise that a project will pay its own way.

What would change our assessment

Shorter connection queues, faster equipment deliveries and completed projects would weaken the bottleneck thesis. So would evidence that rising demand is being served without shifting costs to other customers. Delays, weak utilization and rising unrecovered infrastructure costs would strengthen the concern.

How to read the research

The report is a method demonstration with conditional scenarios. Its historical model scores are not measured national conditions. The small retained market backtest does not establish stock-selection skill. This update also corrects an important point: scenario weights that add to 100% are not valid event probabilities when outcomes and time windows overlap.

Sources and further reading

1 International Energy Agency | Key Questions on Energy and AI | April 2026

2 Virginia SCC | Data Center Initiatives | February 2026

Companion research: The AI Data-Center Build-Out - September 26, 2026 revision. Earlier model results remain historical; this evidence refresh does not represent a new model run.